POLITECNICO DI TORINO

Corso di laurea in Ingegneria Meccanica (Mechanical Engineering)

Class L-9 (DM270)

Bachelor’s thesis degree

CAD Modelling and CFD study of a Fluid Controls dual pressure relief valve

Tutor

Prof. Roberto Finesso

Co-Tutor

Prof. Massimo Rundo

Candidate

Resion Emini

July 2024

Acknowledgments

I would like to express my deep and sincere gratitude to my thesis supervisor, Professor Roberto Finesso, for giving me the opportunity to do the thesis with him and providing invaluable guidance throughout this topic. His dynamism, vision, sincerity and motivation have deeply inspired me. I say this because he taught the Fluid Machinery course, which was the course I enjoyed most during my bachelor's degree. It was a great privilege and honor to work and study under his guidance. I would also like to thank him for his friendship, empathy, and great support.

I am extremely grateful to my family for their love, sacrifices for educating and preparing me for my future. I am very thankful to my mum and my sister and my partner for their continuing support to complete the bachelor's degree. They have been close in every moment, always reminding me of my potential.

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Introduction

Valve is a highly spread component used in many industries and applications. AS the valve discussed in this thesis is a pressure valve, there are two fundamental approaches to the design and analysis of pressure control valves, and of systems that involve fluid flow in general. They are experimentation and calculation. The former typically involves construction of models that need to be tested, while the latter involves the solution of differential equations, either analytically or computationally. Computational fluid dynamics (CFD) is the eld of study devoted solving the equations of fluid flow using a computer. Nowadays, both experimental and CFD analysis are applied, and the two complement each other. For example, we may obtain global properties, such as pressure drop and power, experimentally, but use CFD to obtain details about the ow eld, such as velocity and pressure pro les, and ow streamlines. In addition, experimental data are often used to validate CFD solutions by matching the determined global quantities. CFD is therefore employed in sectors such as Fluid Power to shorten the design cycle through carefully controlled parametric studies, thereby reducing the required amount of experimental testing.

This thesis regards the 1L22 dual pressure relief valve, a safety device used in fluid power systems. Initially, after a brief foreword on pressure relief valves in general, the 3D model of the valve is discussed. Particular care is taken in distinguishing the components of the valve providing detailed picture and explanation of the model, since it is fundamental to properly understand how the valve behaves inside a fluid power circuit and what are its operating principles, which are then examined.

In the last chapter, computational fluid dynamics is presented, and the study of the valve is carried out through several simulations. Finally, the importance of mesh independence to obtain realistic and meaningful results is underlined.

The software exploited for the modelling and CFD study of the valve is SolidWorks, a widely used parametric-based software produced by the Fluid Systsms,inc Company.

Chapter 1

Pressure relief valves

Fluid systems are fluid circuits used to transport fluid from a location to another or to generate, control, and transmit mechanical power from a prime mover (which can be an electric motor, an internal combustion engine or a boiler) to a user to perform work. Fluid power systems have a wide variety of use because of a large number of advantages such as transmission of a high power over a greater distance than mechanical systems in a more economical way, they simpler to operate or maintain, and also can multiply forces from a fraction to a considerable value due to difference in cross section of the fluid flow. Of course, except of advantages, like every other power generating system, fluid power systems have their own disadvantages. As oil is now preferred as lubricant due to its properties and because it does not rust the metals, still leakage is impossible to completely eliminate. Another reason is that of producing high hydraulic power inside a fluid power system. Considering constant volumetric flow, from Conservation of Mechanical Energy equation with a control volume (CV) to analyse around a pump, we have:

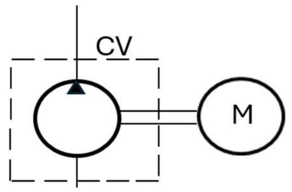


Figure 1.1: Control volume around a pump

$$P = Qp_2 - Qp_1 + P_w$$

Where P is the power required to the pump, $Qp_2 - Qp_2$ is the hydraulic power produced and P_w expresses the fluid losses.

As we see, higher the pressure at the outlet of the pump, higher will be the hydraulic power produced. High pressure can be dangerous and cause injuries to people, so for this reason, we use a safety device to control overpressure condition. An overpressure condition refers to any condition which would cause pressure in a vessel, system, pipe, or storage tank to increase beyond the design pressure of the valve or maximum allowable working pressure (MAWP) On this thesis, we will discuss one kind of safety valve called Pressure Relief Valve (PRV).

The first safety valve (Figure 1.2) was invented in 1681 by Denis Papin, who was born in Blois, France, in 1647. He commenced his experiments on the phenomena of steam in July 1676, at London, under Robert Boyle (the person who discovered Boyle's law).

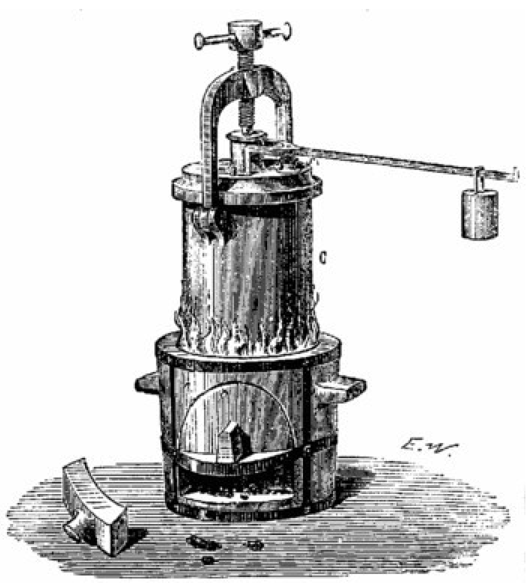


Figure 1.2: First design of a pressure relief valve

High-pressure steam boilers were blowing up across the land and a big reason for that had to do with the relief valves of the time. These were longish levers with a pivot point on the boiler end, a plug for the release of steam near the pivot point, and an iron weight at the other end of the lever to keep the plug-in place until it needed to pop.

The operating engineer could control the relief pressure of this crude device by moving the weight further out on the lever, or by adding more weight to the lever if the spirit moved him.

Denis Pupin idea worked, but the problems were, to sustain high pressure, his valve needed large space and it used to occur premature opening of the valve if the device was subjected to bouncing movement.

To avoid these disadvantages of the Denis Papin's valve, direct-acting deadweight pressure relief valves were installed on early steam locomotives. But also in this case, we had premature opening due to bouncing movements as well as very often explosions due to pressure stuck inside the vent of the locomotives (locomotives were not able to release all the excess pressure.

Then, around 1828, Timothy Hackworth used for the first-time direct acting spring valve on his locomotive engine called the Royal George. A main difference from nowadays models is that we use a coil spring instead of a leaf spring that was firstly chosen.

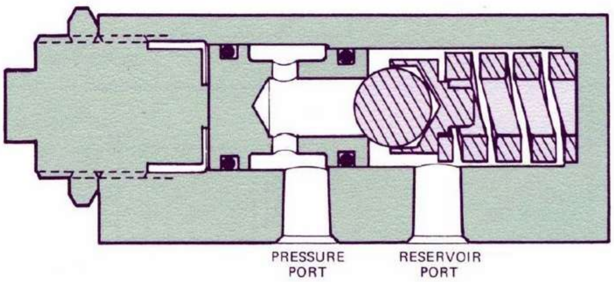


Figure 1.3: Single pressure relief valve

Figure 1.3 shows an example of a single pressure relief valve, and starting from this, we will discuss the working principle and importance of a dual pressure relief valve. This valve is normally closed. The fluid does not flow from the pressure port to the reservoir port. We can see a screw with a nut on the left side, a spring and a ball poppet on the right side and valve is connected with a pressure port where it receives a pressure information, and a reservoir port to discharge the flow when the valve opens.

Chapter 2

1L22 Dual pressure relief valve and components

1L22 dual pressure relief valve is a direct acting valve which body's is made of hi-strength wrought aluminium alloy and internal parts of hardened steel. The fact that is 'dual', it works in two directions, so it increases the number of applications in which we can adapt this device. 1L22 is used in many cases to relief shock pressures at control valves and actuators when controls are suddenly centred or reversed. Also relieves shocks when external loads are applied to the actuators. The design makes possible to work at a rated flow of 40 gallons per minute (G.P.M) and at a working pressure of 5000 pounds per square inch (P.S.I) or 350 bar.

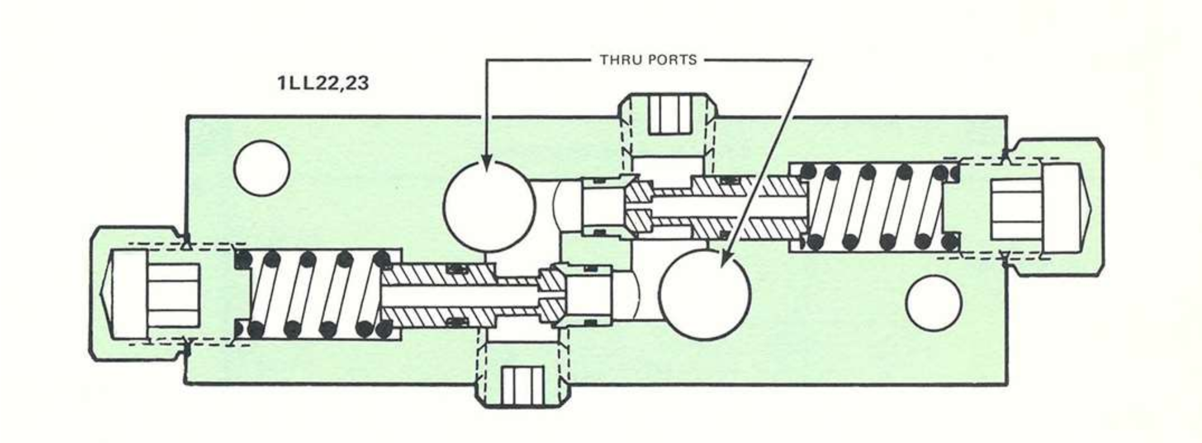


Figure 2.1: 1L22 Dual Pressure Relief Valve

Thanks to SolidWorks 3D, it was possible to fully design the external casing of the valve, every internal component and the assembly of all parts. All the dimensions were taken from a model available on ‘Fluid Controls, inc’.

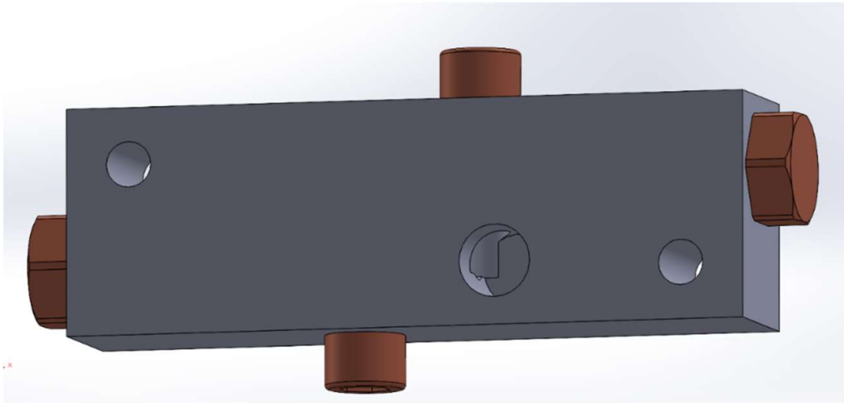


Figure 2.2: Isometric view of 1L22 dual pressure relief valve

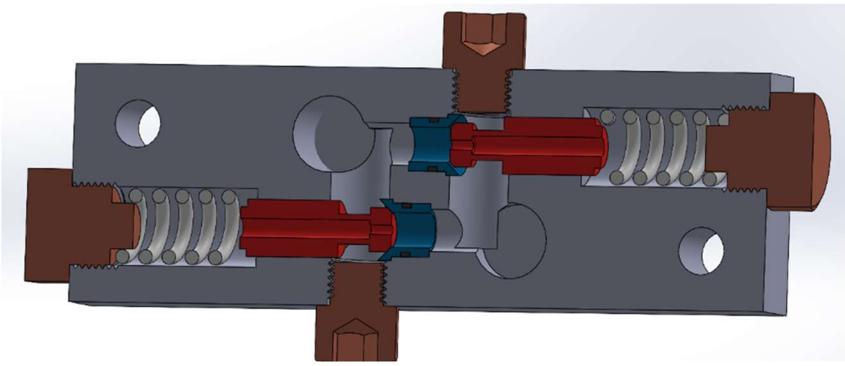


Figure 2.3: Cross section view of 1L22 dual pressure relief valve

Initially the external casing (Fig. 2.4 and Fig 2.5) was modelled with all the needed channels and the threads for the screws taken by ISO standards. The two corner small-diameter free holes are for the purpose of mounting the valve, while two other free holes with the bigger diameter are for the channels of fluid to flow.

Two drilled holes on the vertical position are for the purpose of creating the fluid channels which are closed by two Allen head screws while the two drilled holes on the horizontal position are for the screw to adjust the preload of the spring. The preload of the spring together with the red part on Fig.2.3, which is called poppet, will define the maximum pressure of the system.

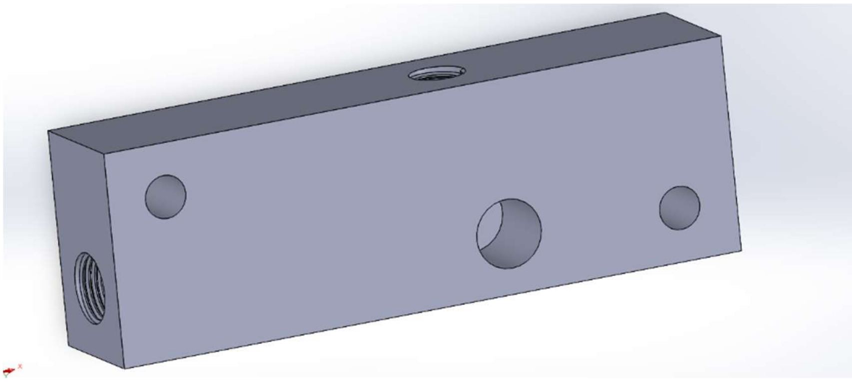


Figure 2.4: External casing of the valve; isometric section view

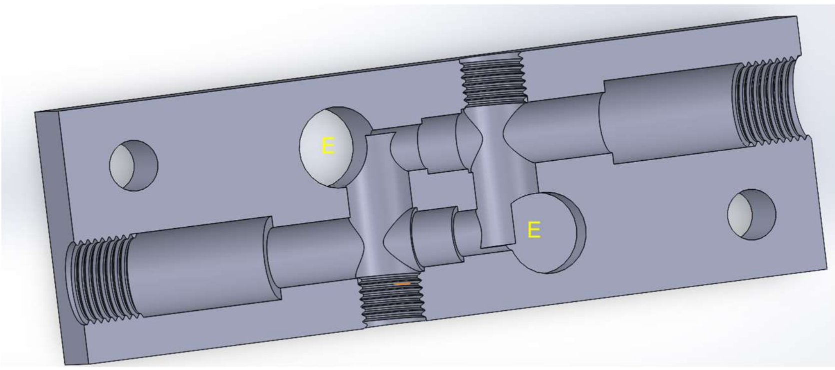
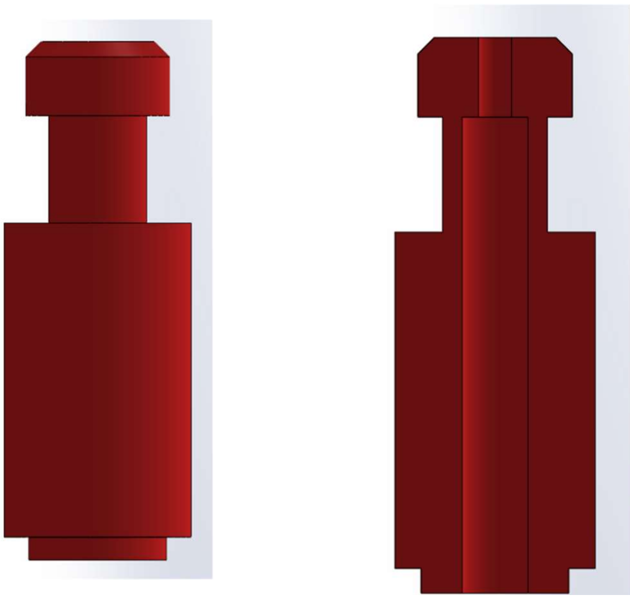


Figure 2.5: External casing of the valve; cross section view

The fluid enters on one of the ports named “E” (“E” stands for entry or exit) and exits on the other port again named “E”. The detailed procedure how fluid moves inside the valve will be explained on the next pages.

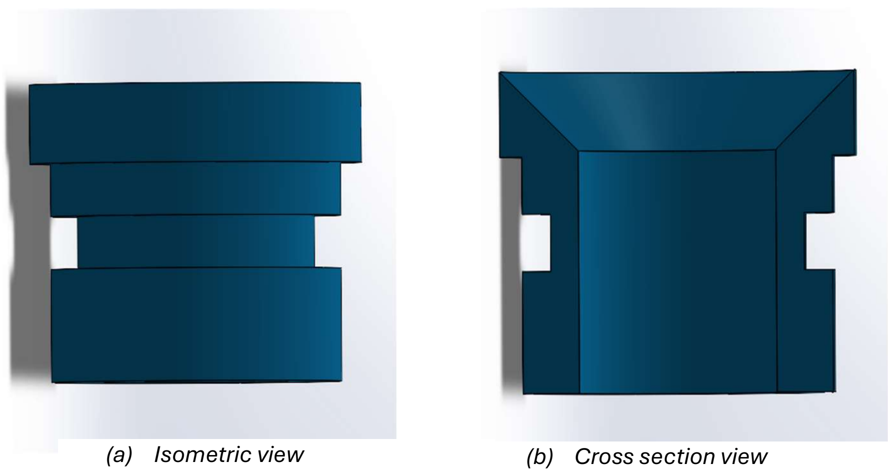
Poppet and sleeve are very general parts of the valve. Poppet is the element connected with the spring and with the sleeve on the conical part. Spring keeps the poppet attached to the conical part of the sleeve, which mechanism block the fluid flow.

The empty hollow cylinder missing on the sleeve is for the purpose of attaching an oil sealing element in the shape of the ring to seal the sleeve and the valve.



(a) Isometric view *(b) Cross section view*

Figure 2.6: Poppet



(a) Isometric view *(b) Cross section view*

Figure 2.7: Sleeve

Chapter 3

Valve operations

3.1 Operation of a pressure relief valve inside a circuit

Fluid power systems are shown by graphical representations as shown in Fig. 3.1. Each component is represented by a symbol, at rest position. In this way we can explain the function of the fluid power system, operating condition, and connection between the different components of the circuit.

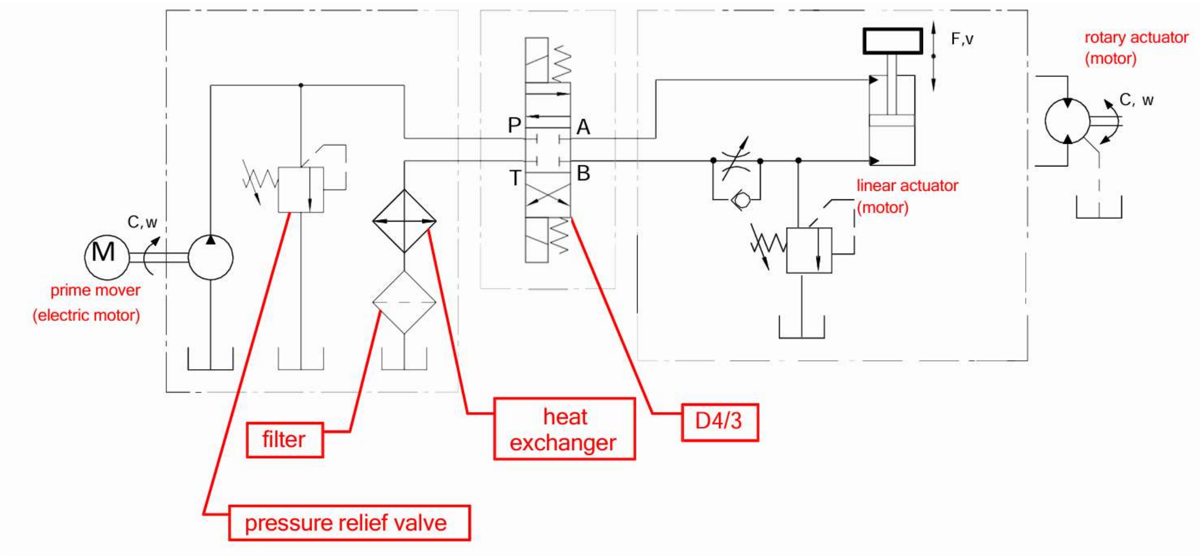


Figure 3.1: Example of a fluid power system

The symbol of a pressure relief valve is shown in Fig. 3.12. The fluid enters at point P and exits at point T when the valve is opened. The square represents an operating condition of a valve. Pressure relief valve is normally closed. To transfer pressure information, pilot lines are used. Fluid flows through pilot line and exerts a pressure at surface S. That hydraulic force is balanced by the spring which create a force with an opposite direction. When the hydraulic force is higher than the spring force, cracking pressure is achieved, vertical arrows shift to the left, it aligns with the main tube, and fluid discharges to the tank.

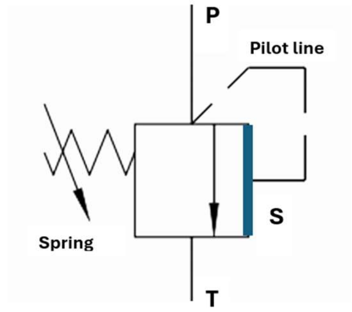


Figure 3.12: Pressure relief valve symbol

The arrow shown to the spring means that force that the spring can exert can be regulated by the user. So, when cracking pressure is reached, we have the condition:

$$F_S = p^* S \qquad\qquad p^* = \frac{F_S}{\frac{\pi D^2}{4}}$$

Where F_S is the force exerted by the spring and p^* is the cracking pressure, D is the diameter shown in the Fig. 3.13.

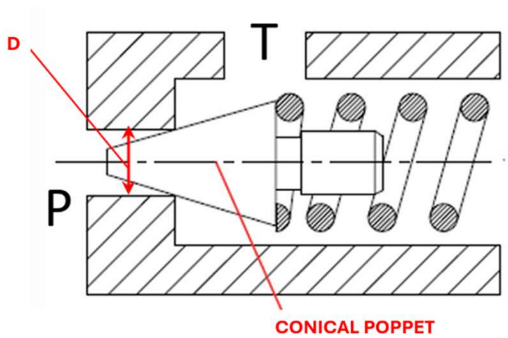


Figure 3.13: Single stage pressure relief valve

3.2 Behaviour of 1L22 inside a circuit

The way 1L22 dual pressure relief valve can be explained is through an application as shown in Fig. 3.4. The elements on this application are the pump, single pressure relief valve on the top left side, a direct control valve in the middle, dual pressure relief valve and the motor. Direct control valve is used to direct the flow and it has three operating conditions, parallel flow (upper operating condition), crossed arrow flow (bottom operating condition) and rest position. Motor can rotate on both, clockwise and counterclockwise directions based on the operating condition of the directional control valve.

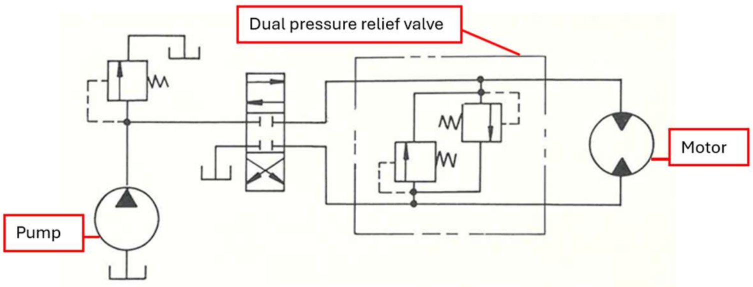


Figure 3.21: Dual pressure relief valve application

If the directional control valve is working in a parallel flow configuration, the fluid will flow through the upper line, it will rotate the motor on one direction and the flow rate is discharged by the bottom line. The upper line is the high-pressure line while the lower side is the low-pressure line, and we want the dual pressure relief valve to limit the pressure on the high-pressure line which is the upper line.

For example, on the Fig. 3.22, the fluid is flowing on a parallel flow configuration, so initially on the upper line. If the pressure of the flow is exceeding the maximum pressure limited by the valve, the valve opens, fluid flows on the bottom line and it discharges the flow to the tank.

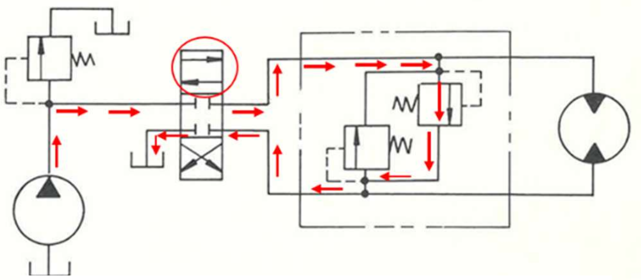


Figure 3.22: Path of the fluid in the circuit

The other configuration that the circuit can work is that of crossed arrow positioning. Now the flow rate, so the hydraulic power is delivered to the bottom line, which in this case is the high-pressure line. The upper line is connected to the tank, so we have the low pressure on the upper one. We need another pressure relief valve to limit the pressure on the bottom line.

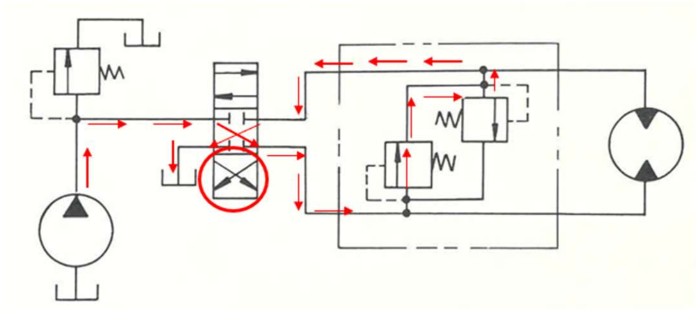


Figure 3.23: Crossed arrow configuration of the valve

Meanwhile what happens inside the valve is that the fluid enters in one of the channels (like in the Fig 3.23), exerts a force on the poppet, and this force is counterbalanced by the spring when the valve is closed, therefore the force exerted by the pressure is less than the force of the spring.

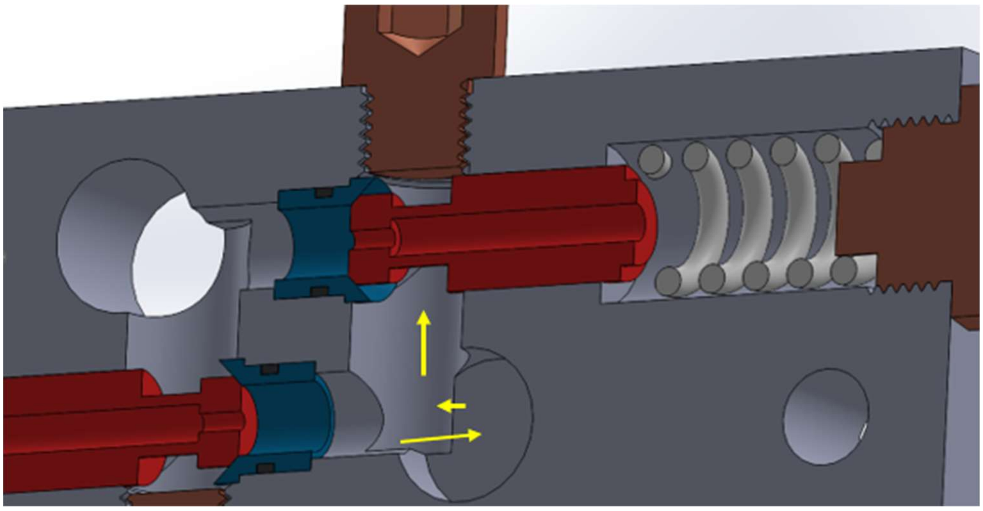


Figure 3.24: Fluid flow from 1 port

Even though fluid pushes the poppet on both sides, area in which the pressure acts an opening force F2 is larger than area of the closing force F1, therefore we will have a resultant opening force Fr acting on an opposite way of force of the spring Fs, pushing the spring to compression state and moving the poppet to the right (Fig. 3.26) when Fr = Fs. This displacement of the poppet will make the fluid flow through the other port and discharge to the tank.

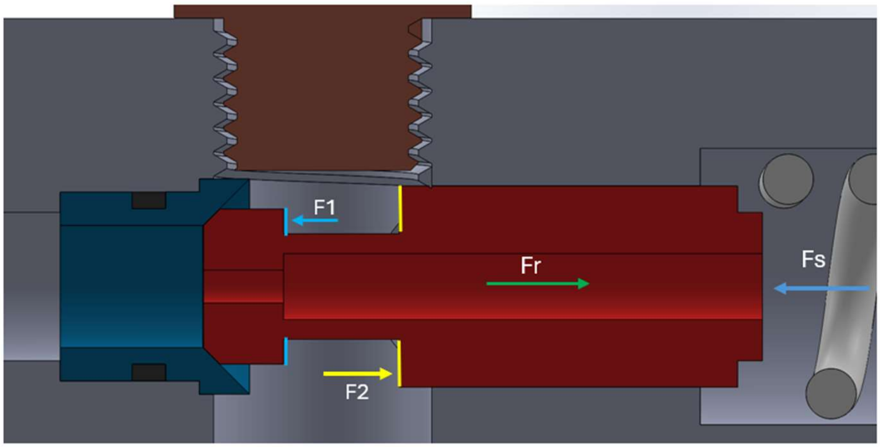


Figure 3.25: Force balance on the poppet

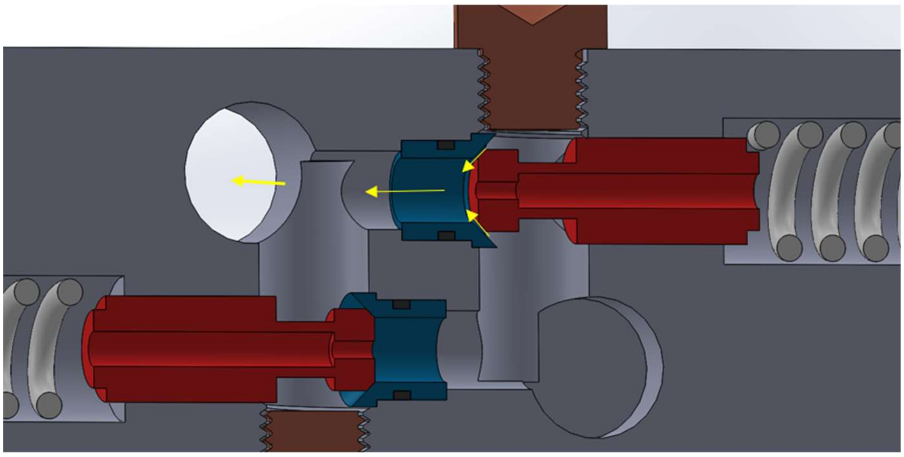


Figure 3.26: Outer flow from the valve

3.3 The benefits and improvements for dual pressure relief valve

Normally, in a circuit in we would need one pressure relief valve to limit the pressure on the upper line when the direct control valve is working in parallel arrow configuration, and another pressure relief valve for the bottom line, when direct control valve is working in crossed arrow configuration as shown in Fig.3.31. This solution would require more space as we would need to connect two valves with tanks using an external piping.

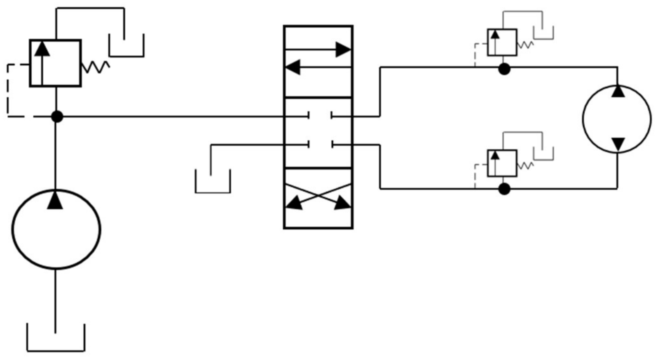


Figure 3.31: Circuit with two pressure relief valves

An improvement could be adding a pilot stage. We will see now why adding a pilot stage would improve the functioning of the valve. A pilot stage is a smaller valve which is connected with the main valve and helps the main valve to achieve higher accuracy, reliability and efficiency.

Both solutions would require more space for the system, so we can use instead a more compact and cheaper solution, which already includes the two pressure relief valves, and which can do the same function with a single casing. This could help us to save space and to make the circuit simpler.

As we said, a solution which would improve the working of the valve would be adding a pilot stage. Let me explain why. In a single stage pressure relief valve, we have the force equilibrium:

$$F_S = p^* S \qquad\qquad p^* = \frac{F_S}{S} = \frac{F_0 + k_p x}{S}$$

where x is the displacement of the poppet. Since x is not constant during the opening of the poppet, the force exerted by the spring and the cracking pressure are dependent on the displacement x.

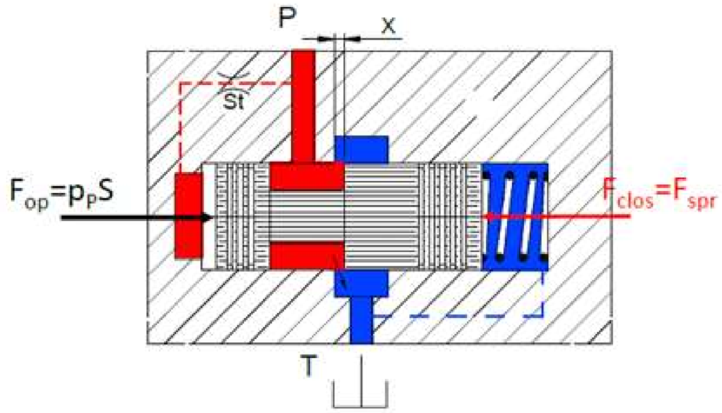


Figure 3.32: Single stage pressure relief valve

As the discharged flow rate increases, the spring get compressed, therefore the pressure increases non-linearly (term used is “over-ride”). The pilot stage can be adapted with a small pilot poppet and a very stiff pilot spring to properly set the cracking pressure. The main poppet has a relatively large dimension and a spring with low stiffness. Hence, the main stage cracking pressure will correspond closely to the pilot stage value (Fig 3.33). This can be set by adjusting the spring pre-load on the pilot stage, and it will be kept nearly constant, with no considerable override.

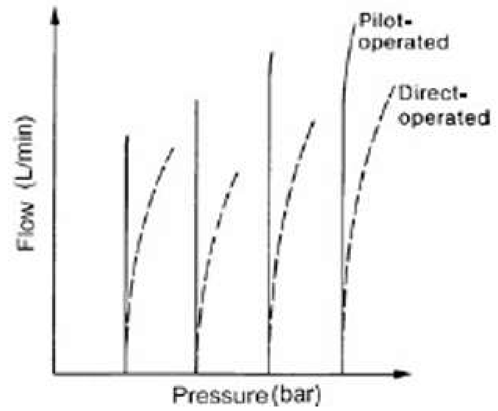


Figure 3.33: Flow vs pressure for different p*

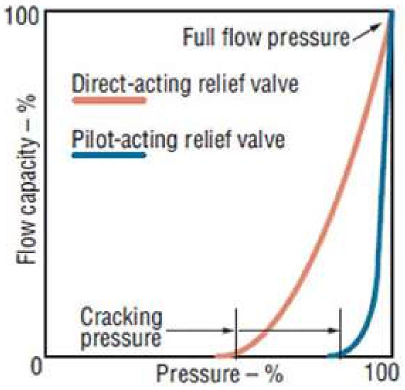


Figure 3.35: Flow capacity % vs pressure %

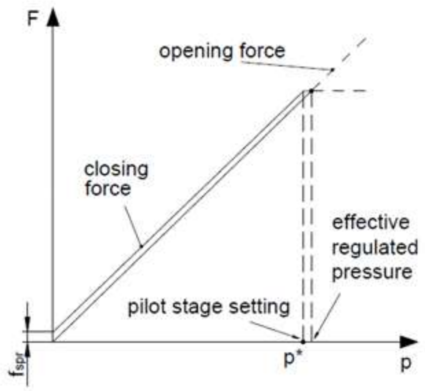


Figure 3.34: Forces vs. pressure graph for the main stage of a dual stage relief valve

3.4 Flow through orifice and the discharge coefficient

When the valve is opened (in regulating condition), at the conical surface of the fluid flow (next to the poppet), we have formation of an orifice. An orifice is the restriction of the area inside a pipe. As we see on the Fig.3.41 the fluid enters with a velocity v1, and there it faces the boundary of the orifice, and then at the section 3 we have again the velocity v1. Downstream the restricted section there is a further restriction of the area which generate the so called '*vena contracta',* that is the minimum section for the flow.

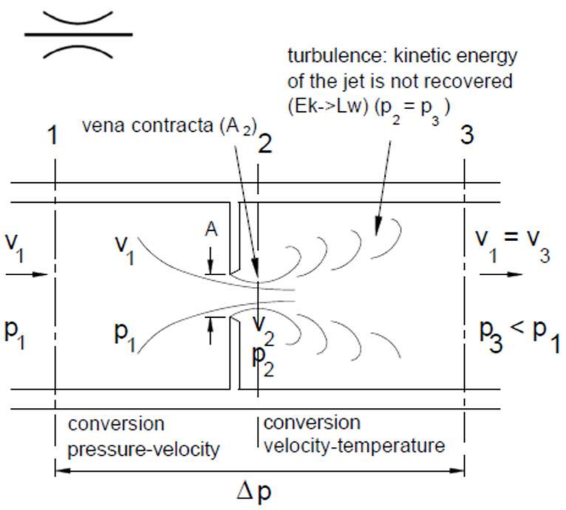


Figure 3.41: Vena contracta and turbulent flow through an orifice

The cross section of *vena contracta* is estimated by multiplying the geometrical area A by a coefficient C_c that is the contraction area coefficient. From the conservation of mechanical energy, when the fluid accelerates in the *vena contracta*, a part of the pressure is converted into kinetic energy, therefore we have a reduction in pressure. Downstream near section 3, unfortunately we do not have a conversion of kinetic energy into pressure again, because we have an expansion of the flow and formation of many vortices, which generates a lot of viscous dissipations and therefore the pressure is not recovered. So, on section 3 pressure p_3 is lower than p_1.

How can we estimate the flow rate through this restrictor?

Conservation of mechanical energy:

$$L_i = \int_1^2 v\,\partial p + \Delta E_k + \Delta E_g + \Delta E_\omega + L_w = 0 \qquad (3.1)$$

Considering an *isentropic evolution* from section 1 to 2 since losses are very small, we obtain from Eq.3.1:

$$\frac{p_2 - p_1}{\rho} + \frac{v_{2,is}^2 - v_1^2}{2} = 0 \qquad (3.2)$$

From Eq.3.2, considering $\Delta p = p_1 - p_2$, we have:

$$v_{2,is} = \frac{\sqrt{2\frac{\Delta p}{\rho}}}{\sqrt{1-\frac{v_1^2}{v_{2,is}^2}}} \tag{3.3}$$

c_v is called *velocity coefficient* and is an empirical factor that takes into account the effect of viscous friction.

Continuity equation for incompressible fluid:

$$Q = A_1 v_1 = A_2 v_2 = A_2' v_{2,is} \tag{3.4}$$

where c_v is called *velocity coefficient* and is an empirical factor that takes into account the effect of viscous friction. It has a value approximately equal to 0.98, but it is usually considered as one for most calculations. Therefore:

$$\frac{v_1}{v_2} = \frac{A_2'}{A_1} \approx \frac{A_2}{A_1} = \frac{C_C A}{A_1} \tag{3.5}$$

We can write again the equation 3.3 in terms of areas:

$$v_{2,is} = \frac{\sqrt{2\frac{\Delta p}{\rho}}}{\sqrt{1-\left(\frac{C_C}{A_1}\right)}} \tag{3.6}$$

Finally, it is possible to see that the volume flow rate Q depends on the pressure on the entrance between inlet and outlet, and on the flow area of the orifice:

$$Q = A_2 v_2 = (C_C A)(c_v v_2) = C_C A c_v \frac{\sqrt{2\frac{\Delta p}{\rho}}}{\sqrt{1-\left(\frac{C_C A}{A_1}\right)^2}}$$

$$Q = C_e A \sqrt{2\frac{\Delta p}{\rho}} \qquad where \qquad C_e = \frac{C_C c_v}{\sqrt{1-\left(\frac{C_v A}{A_1}\right)^2}} \tag{3.7}$$

In the last equation, C_e orifice *discharge coefficient* or *flow coefficient* and usually it takes value in the range [0,6; 0,9]. It depends on contraction area coefficient C_C, on velocity coefficient c_v, on the area A, and on the area of the pipe. In general, it is not constant.

The discharge coefficient is discovered that if it is plotted as a function of Raynolds number, it is a function of just Raynolds number (Fig. 3.42).

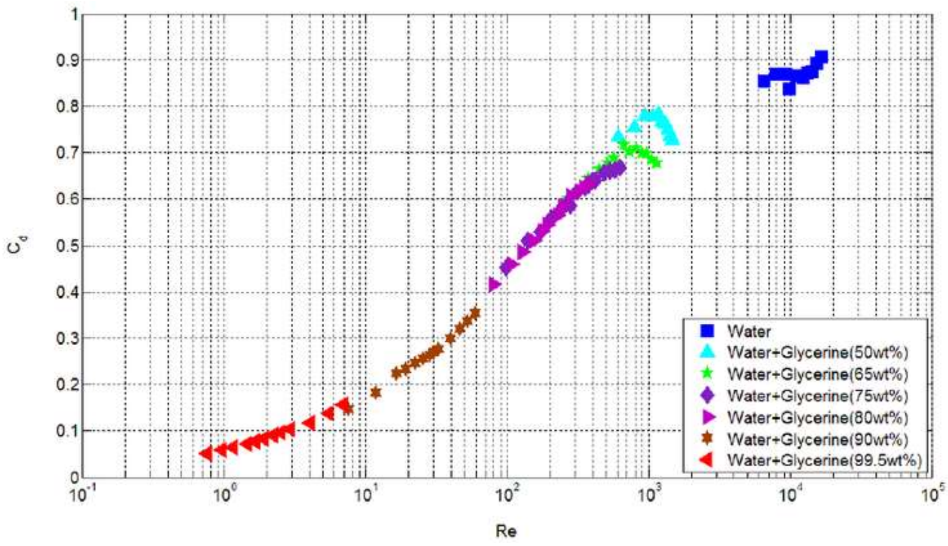


Figure 3.42: Discharge coefficient versus Reynolds number for different test liquid

On this graph, we can see that for low Reynolds numbers (usually in laminar flow), the discharge coefficient varies linearly, while instead for high Reynolds number, it has a constant value between 0.8 and 0.9. This is explained by the fact that viscous losses are more significant at low Raynolds numbers. Having a low Reynolds number means viscous forces are higher than inertial forces, which keeps the flow in a laminar path.

Chapter 4

CFD Simulation

4.1 CFD meaning and equations

CFD is a computational fluid dynamic numerical technique which allows to simulate fluid dynamics inside the valve, and it is based on *Reynolds-Averaged Navier-Stokes (RANS) equations*, which are a set of equations including conservation of the mass, of the momentum and of the energy for the system.

1. Mass is conserved.
2. Newton's second law, F= ma.
3. Energy is conserved.

We take these three equations, and we apply them to a suitable model of the flow. Applying a suitable model of the flow means describing the volume size, shape, and the dynamic (moving or static volume). In this way we can extract the right mathematical equations to be solved.

The volume is discretised in several sub volumes which are called mesh (Fig.4.1). So, the equations are discretised and solved for each individual element of the mesh. In the end, we have an algebraic set of equations, one equation for each sub volume. The solution is the field of pressure, temperature and velocity of the fluid. We are interested in pressure and velocity and not in temperature, since we are not making a heat transfer simulation and we are not interested in heat transfer effects.

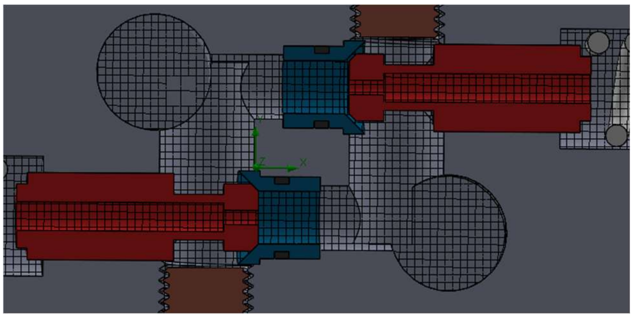


Figure 4.1: Global mesh

If we write *Navier-Stokes equations*, we have:

$$\rho \frac{D\vec{v}}{Dt} = \mu \nabla^2 \vec{v} - \nabla p + \rho \vec{g}$$

This coupled system of nonlinear partial differential equations is usually very hard to solve analytically. Using computational fluid dynamics, we can analyse a fluid problem by following these steps.

1. Firstly, we need to write the mathematical equations describing the fluid flow conditions.
2. Then, we discretise these equations to produce a numerical analogue of the equations (a numerical method to solve them).
3. The domain is divided into small grids. The quality of a CFD simulation is highly dependent on the quality of the mesh.
4. Finally, the initial conditions and the boundary conditions of the specific problem, which can be the pressure, or the flow rate. The solution method can be direct or iterative. In addition, certain control parameters are used to control the convergence, stability, and accuracy of the method
5. Once the solution has converged, flow field variables such as velocity and pressure can be plotted and analysed graphically.

Simulation that is performed using SolidWorks flow simulation tool is a static one. We place the elements in the working positions. In this case the metering (poppet) will be positioned in the working position. Solution will be the steady state solution. The quality of the result as is written before depends on meshing so the size of the cells. The smaller the size, so the more refined the mesh, the more accurate the results (as we see in Fig. 4.11 and 4.12). The problem is if we increase the number of cells, we increase the number of equations to be solved, so we increase the computational time. That's why we only refine the mesh in correspondence with the regions which are more critical, which we have the largest gradient of pressure and velocity around the restrictor in our case.

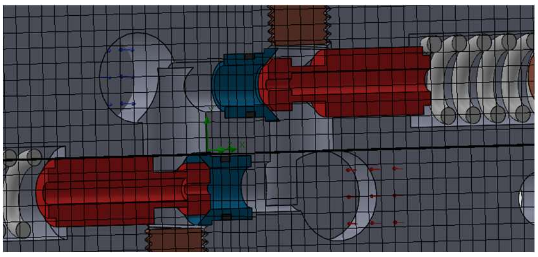


Figure 4.11: Initial global mesh

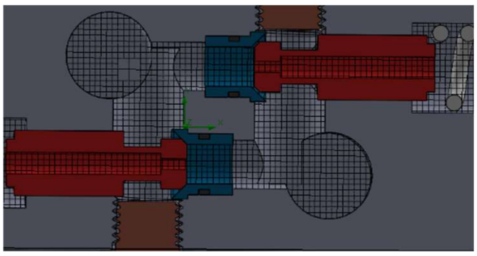


Figure 4.12: Increased number of cells

How can we understand how much the mesh is refined? We do a grid sensitivity study; we start from a condition without any local refinement (also called global mesh). Then, we progressively add a local mesh and progressively refine this local mesh. After each simulation we look at the result to see if the results continue to change when we refine the mesh. When the results won't change anymore, we have achieved the grid independence, and the result does not depend anymore on the mesh. We calculate the flow coefficient across the restrictor from this study. After each simulation we calculate the flow coefficient. If we see the value is not changing anymore, so we achieve the grid independence, we find the value of Ce.

4.2 Working fluid and boundary conditions

To generate the flow, as we said we need to impose a working fluid and the boundary conditions. First thing to do is to generate the ports of fluid flow which is also called the lid. We pick two lids, on one of them an inlet lid where fluid will enter and one for the outlet lid. In the Fig. 4.2 we can see the inlet lid where the fluid enters, and the outlet lid where the fluid exits.

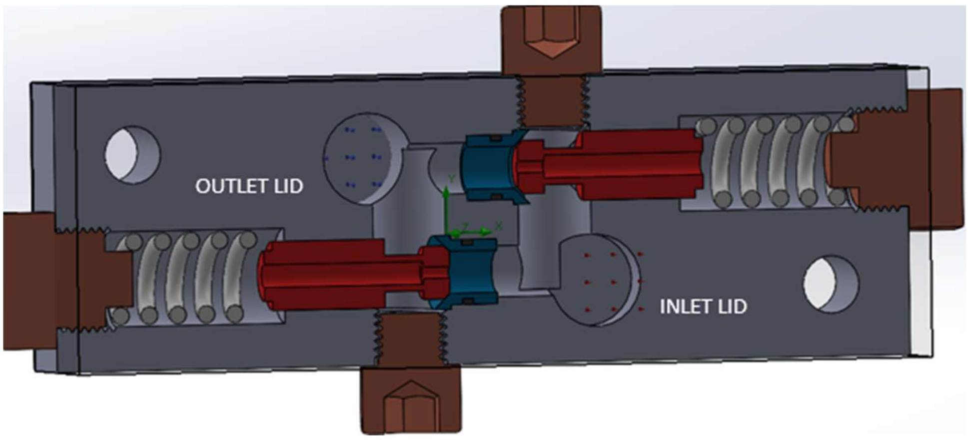


Figure 6: Inlet and outlet lid

With the Check Geometry command, it is possible to verify that the fluid volume is correctly generated and there are no open regions (Fig.4.21). It is also important to check if the computational domain (Fig. 4.22), automatically defined by SolidWorks after the start of a new simulation project, contains entirely the fluid domain.

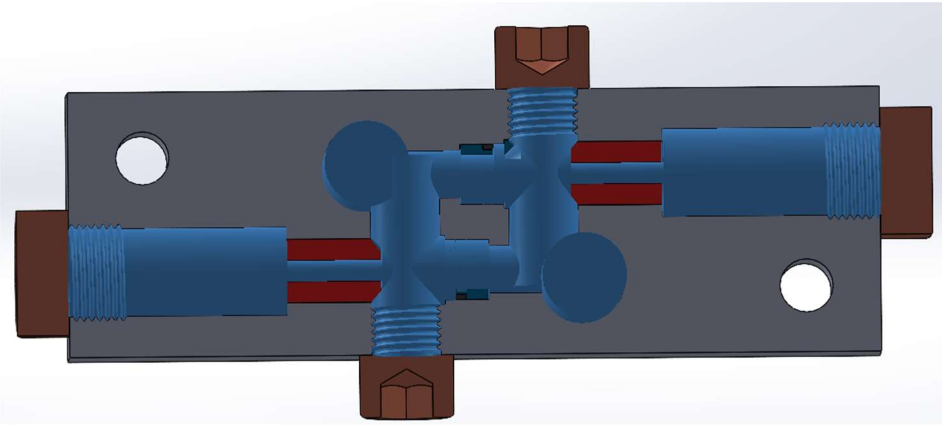


Figure 4.21: Fluid volume

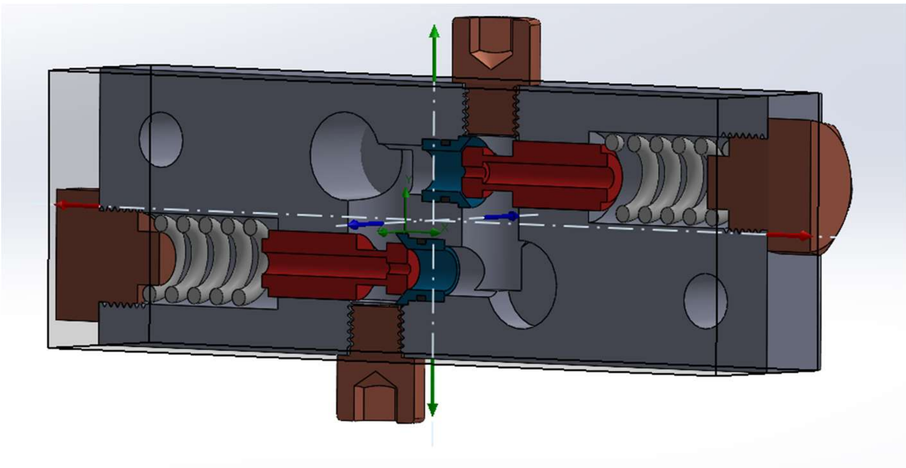


Figure 4.22: Computational domain

The working fluid used to study the valve in the flow simulation is mineral oil. This fluid has a density equal to 850 kg/m^3 and a dynamic viscosity of 0.051 Pa·s. We can add also the heat exchange coefficient or other information regarding the properties of the fluid but they are not in our interest of study.

The boundary condition for the inlet port is picked 100L/min and the outlet boundary is chosen the environment pressure as shown in Fig. 4.23.

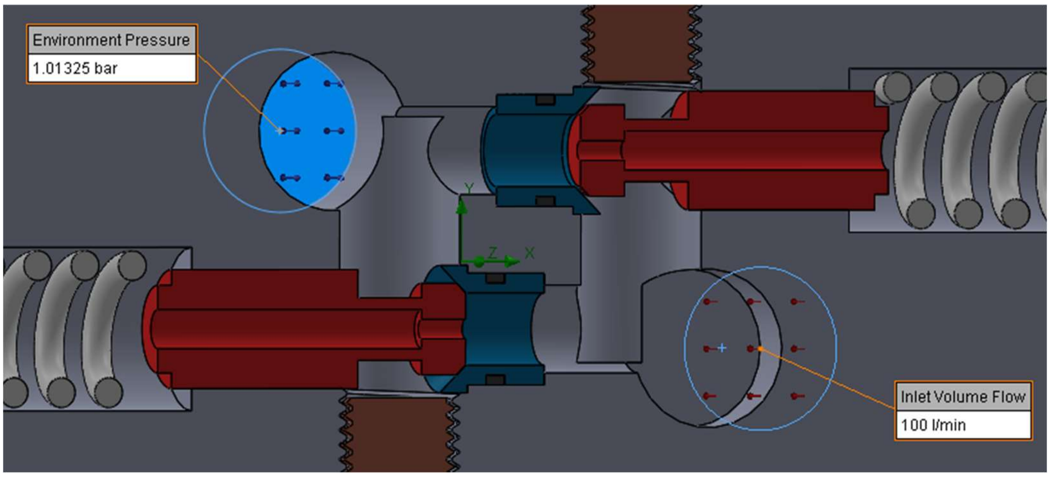


Figure 4.23: Inlet and outlet boundary condition

4.3 Evaluation of the flow areas and poppet positioning

From the set of equations 3.7, we can say that discharge coefficient depends on the flow rate, and the flow rate depends by the area of the metering restrictor. The flow area of conical poppet coincides with the lateral surface of a truncated cone. Its value depends on the seat internal radius r_2 and on the clearance between the seat and the poppet x, as the latter moves in accordance with the inlet pressure.

For our calculations, we will calculate the *lateral area* as:

$$A = \pi(r_2 + r_1) \cdot a \tag{4.2}$$

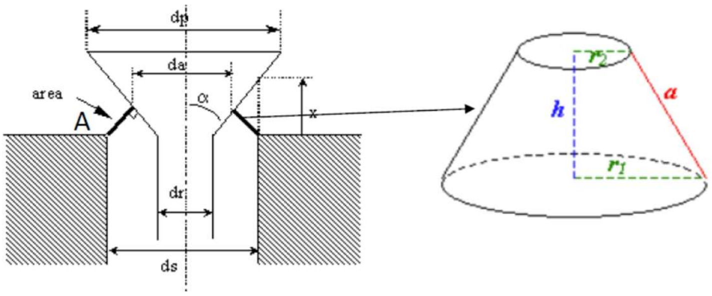


Figure 4.2: Conical flow area

To find the area for positioning the poppet, we can consider two boundary conditions. Volume flow rate of 100l/min at the inlet port and a pressure equal to environmental pressure at the outlet port. We can assume a pressure at the inlet port of 150 bar, density of fluid 850kg/m^3 and an empirical value of discharging coefficient Ce = 0.7.

$$A = \frac{Q}{C_e \cdot \sqrt{2\frac{\Delta p}{\rho}}} = 12.72\, mm^2$$

Now, the next step is to position the poppet such that we obtain an area equal to 12.72 mm^2. This value can be achieved by moving the poppet to the right and calculating r_2, r_1 and a by formula presented on equation 4.2.

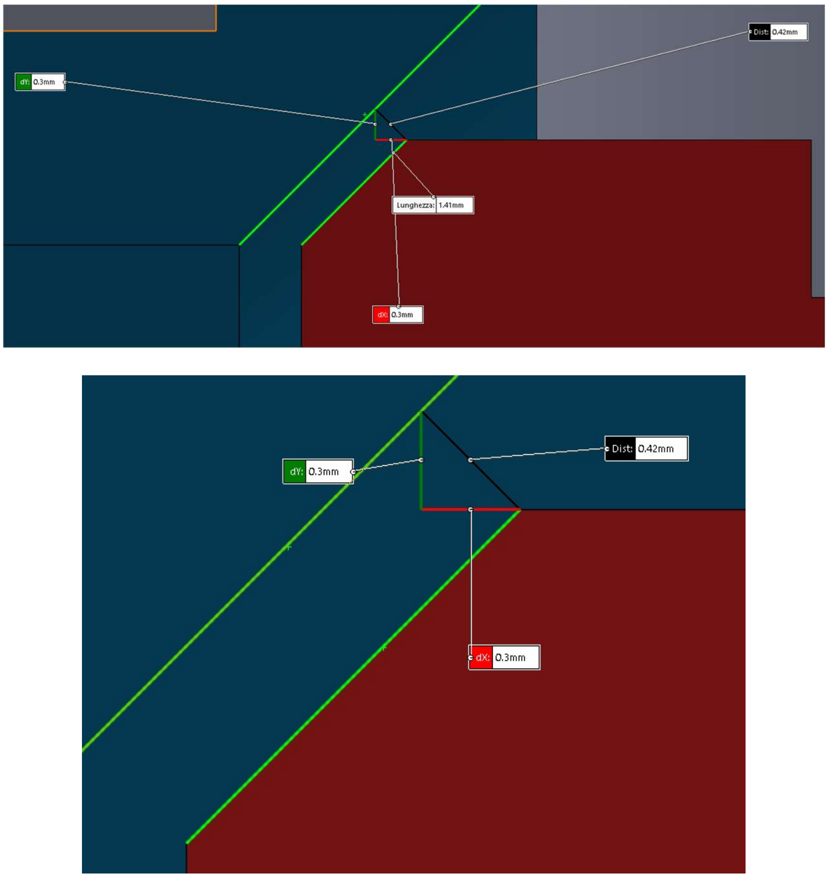


Figure 4.21: Positioning of the poppet

By trying with different value, in the end area was approximated to 12.79mm^2, with r_1 equal to 5mm, r_2 equal to 4.7mm, and a equal to 0.42mm. The next step will be to lunch the simulations and try to estimate the correct value of Ce.

4.4 First simulation

The last step before running the simulation is to generate a mesh over the computational domain (global mesh), for SolidWorks Flow Simulation to be able to solve the equations of motion.

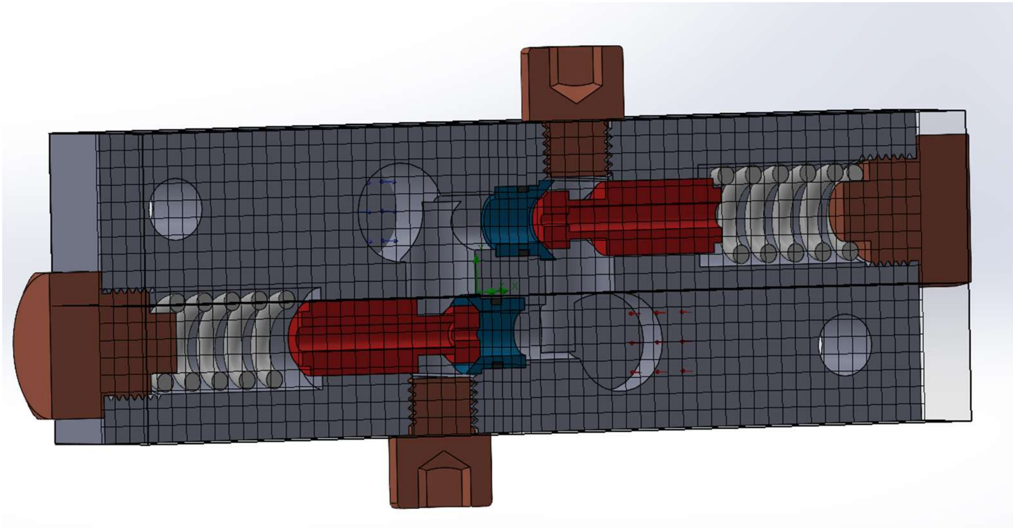


Figure 4.41: General mesh isometric view

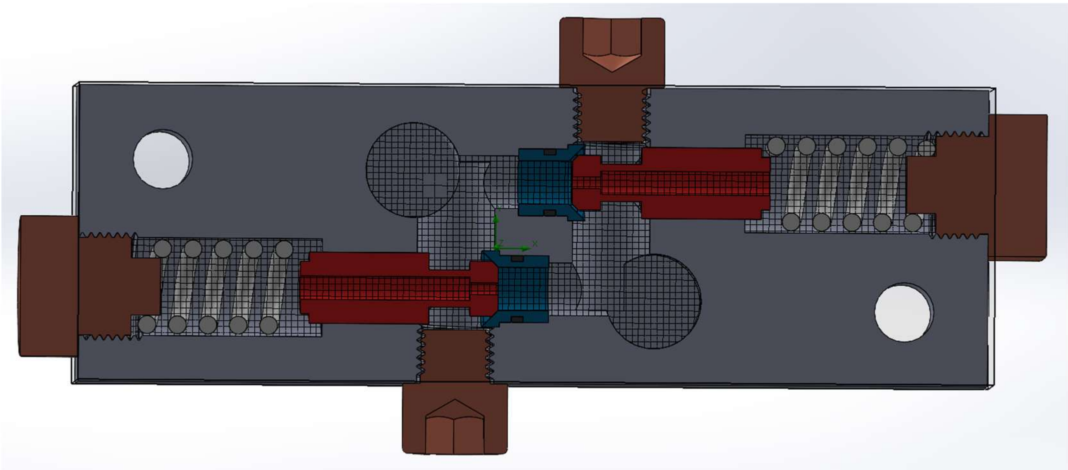


Figure 4.41: General mesh cross section view

SolidWorks generate a global mesh over the whole computational domain (Fig. 4.4 and 4.41). This automatic mesh is identified on a scale from 1 to 8, where the value 1 indicates a coarse mesh while the value 8 indicates a very ne mesh. The simulation time and the quality of the results are strictly related to the selected mesh. A coarser mesh implies very little time for the simulation to be completed, but it is likely to produce wrong results. On the contrary, a finer mesh produces very accurate results, but it requires much more time. Therefore, it is fundamental to and the ideal mesh, for which the compromise between time and reliability of results is acceptable.

The first simulation is carried out with the global mesh set to 6. The calculation time is very short (less than two minutes) and the results can be displayed on the middle plane of the valve (Fig. 4.7). The cut plot shows the pressure distribution of the fluid inside the valve, which is what is needed to calculate the flow coefficients of the two stages, to see if they correspond to the theoretical ones. It is possible to see that a pressure drop of 101.15 bar occurs across the main stage of the valve (Fig. 4.8). Hence, using equation 3.7, the flow coefficient that results from this first simulation is:

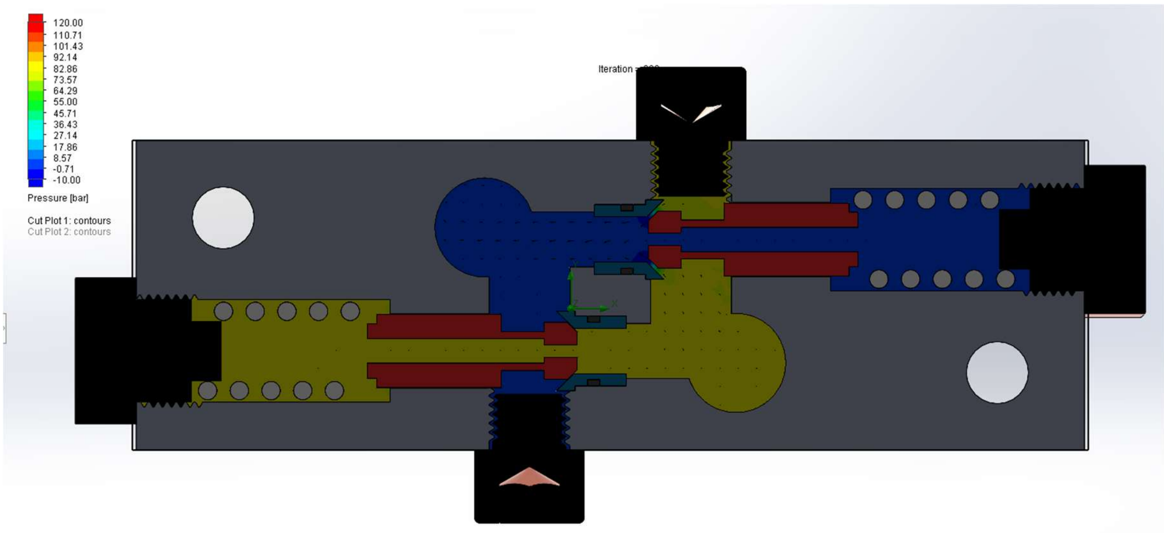


Figure 4.43: Pressure cut plot with global mesh refinement = 6

$$C_e = \frac{Q}{A \cdot \sqrt{2\frac{\Delta p}{\rho}}} = 9.21 \quad (4.4)$$

The value is quiet far from the theoretical one so we can understand that is not accurate. This first simulation is useful to get a general view of the valve behaviour when the main stage is in regulating conditions, but it is necessary to re ne the mesh in order to capture the high pressure and velocity gradients.

4.5 Refined solutions

If we increase the mesh level, we increase the number of equations to be solved, also the simulation time. Since SolidWorks makes a single generic mesh for all the computational domain, it is straightforward to see that the computations of the fluid behaviour in smaller sections is less accurate than the ones in larger areas. To obtain more accurate results in a reasonable amount of time, it is sufficient to refine the mesh in the areas where there is a critical change in cross section, such as the metering restrictor.

Addition of this local mesh correspond in inserting a fictitious solid component in the valve, in the assembly where mesh refinement is needed. This solid component is a simple cylinder geometry with the dimensions of the critical opening area. Since this component is not a real component of the valve but we use it with the purpose to refine the mesh, we must exclude it from the flow domain.

Considering the same pressure cut plot of the first simulation, Fig. 4.5 to 4.53 show the results obtained with the implementation of local meshes, with increasing mesh refinement (starting from 2 and ending with 5)[1]. The pressure distributions are more precise, since the mesh is now able to capture the high pressure and velocity gradients across the metering edges (Fig. 4.13). The flow coefficients can be computed more accurately, as it is shown in the next section (4.6).

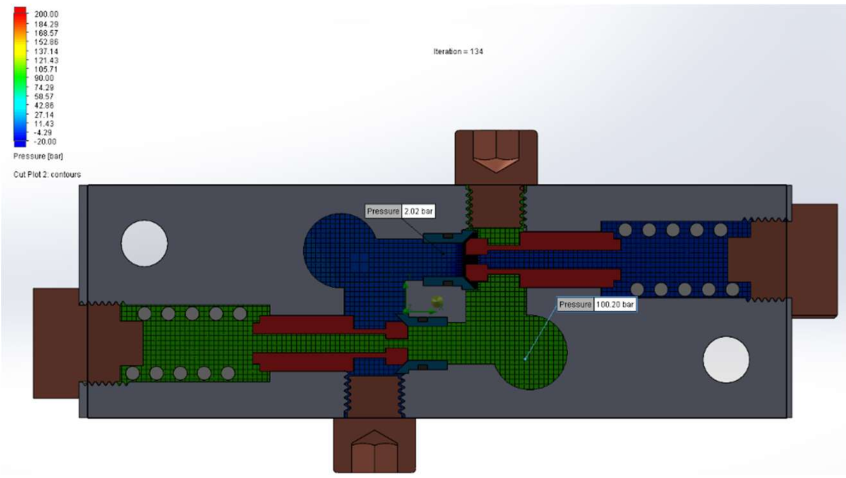


Figure 4.5: Pressure cut plot with local mesh refinement = 2

[1] SolidWorks Flow Simulation does not consider cavitation. This is why the plots show some regions of negative pressure and adopt different pressure scales.

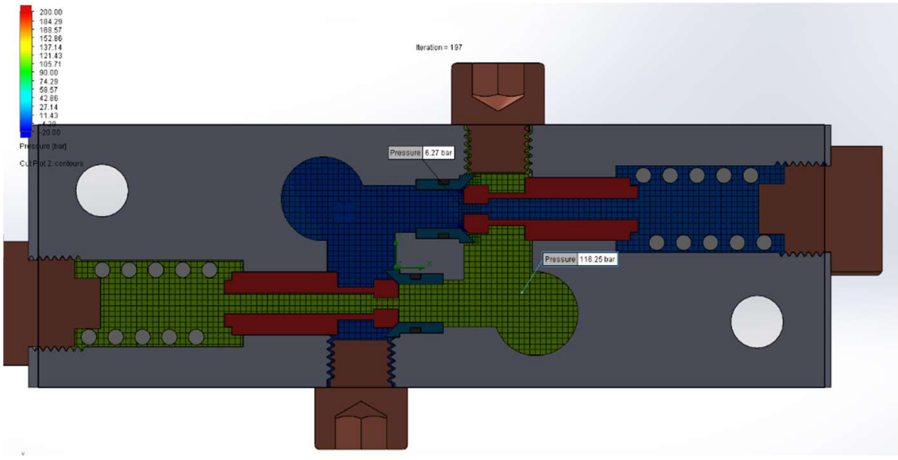


Figure 4.51: Pressure cut plot with local mesh refinement = 3

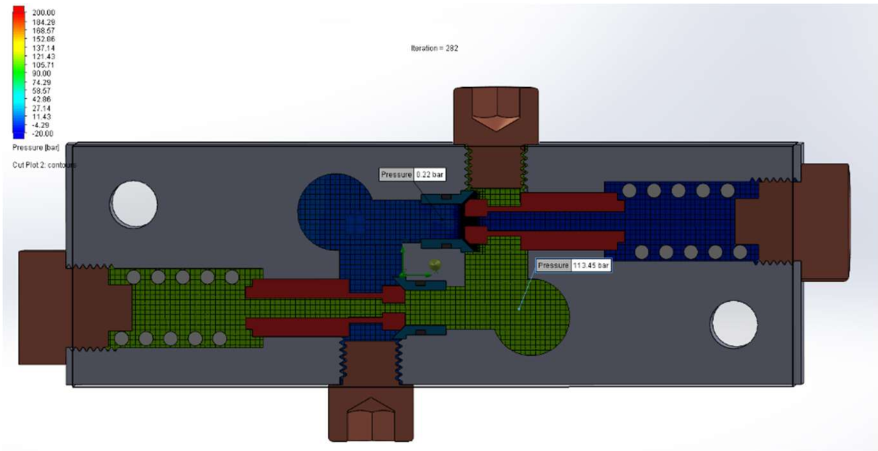


Figure 4.52: Pressure cut plot with local mesh refinement = 4

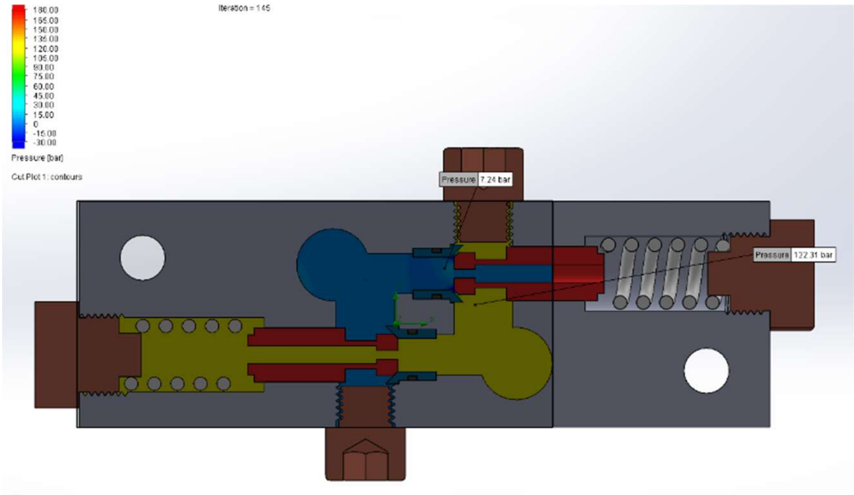


Figure 4.53: Pressure cut plot with local mesh refinement = 5

In the end, stopping with a local mesh refinement was convenient, also because, increasing the local mesh refinement to 6 would require a lot of computations. As we see from the table on the table 4.5 and Fig. 4.54, the number of fluid cells is very high, and so the computational time.

Info

Parameter	Value
Status	Calculation
Total cells	3,210,153
Fluid cells	3,210,153
Fluid cells contacting solids	332,039
Iterations	0

Warning	Comment

Table 4.5: Number of cells

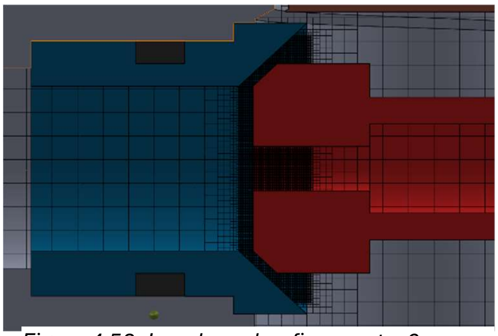


Figure 4.53: Local mesh refinement = 6

4.6 Mesh independence study

In general, a mesh independence study is always needed to be certain that the obtained simulation results are truly acceptable. It allows to find the ideal mesh for the component under analysis, for which computation time and results are reasonable. Each simulation on the 1L22 dual pressure relief valve elaborates a different pressure distribution according to the local refinement level, while the global mesh is kept set to 6 (Fig. 4.5, 4.5 to 4.53). It is possible to analyse how the ow coefficient Ce varies in accordance with the adopted mesh. It can be calculated directly from equation 4.4:

$$C_e = \frac{Q}{A \cdot \sqrt{2\frac{\Delta p}{\rho}}}$$

where area *A* is the flow area calculated on section 4.3 and is kept constant, as well as volume flow rate Q which is a boundary condition equal to 100L/min, and density. Change in pressure is taken from each case as shown in the Fig. 4.5 to 4.53 for each local mesh refinement.

The obtained results are shown in the following graph, in terms of ow coefficient vs. local mesh refinement level (4.6). The pressure differences are taken by means of the *probe tool* available on SolidWorks in points immediately before and after the openings of the two stages, where the pressure gradients are practically constant.

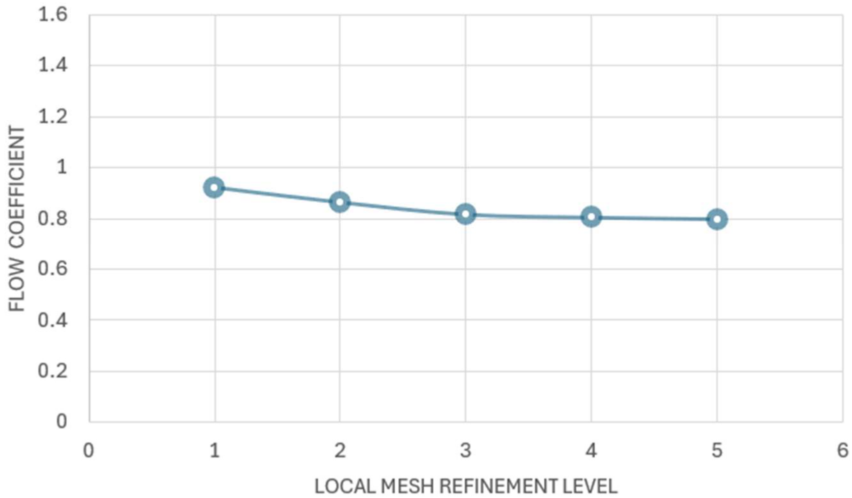


Figure 4.6: Flow coefficient Ce vs local mesh refinement level

It can be observed that as the local mesh gets finer, the computed flow coefficients tend to reach a constant value. This condition is called ***mesh independence*** and indicates the level of refinement from which the ow coefficient has only negligible variations. The graphs show that this is achieved with a local mesh level of 4 for both stages. This value is optimal in terms of computation time and results precision, since the simulation takes about 25 minutes, while it takes up to 1 hour with a level 5 refinement.

Conclusion

The reliability of simulation programs and theoretical models has always played a crucial role in engineering. Prior to the advent of computational fluid dynamics (CFD), designing and studying pressure control valves and systems involving fluid flow relied only on the analytical solution of mathematical equations. This approach often needed numerous hypotheses to simplify calculations, resulting in outcomes that frequently required corrections.

Today, CFD software is utilized to simulate the behaviour of components involving fluid flows, allowing for more precise analyses, and many less errors. SolidWorks, with its *Flow Simulation* extension, exemplifies such software and was employed in this thesis to model the 1L22 dual pressure relief valve and analyse its operating conditions.

1. Initially, the modelling on 3D using SolidWorks was done, and then discussion was focused on pressure relief valves and their importance in fluid power systems, highlighting their important role.
2. Following this, the valve's behaviour within a circuit, its operating principles, and the parameters influencing it were then illustrated.
3. The study then examined the valve under regulating conditions to demonstrate how CFD can enhance the understanding of valve operations. By conducting simulations with progressively finer mesh refinements, it was shown that CFD results are significantly and clearly influenced by mesh quality.
4. Furthermore, it was critical to observe how the flow coefficient *Ce* approached a constant value of about 0.79, getting close to the initial assumptions used to position the two stages. This established the concept of mesh independence, crucial for achieving reliable results.

Upon achieving mesh independence, simulations with varying boundary conditions and different stage positions can be performed. Since the refined mesh is sufficiently detailed to accurately capture high-velocity and pressure gradients in the most critical sections, the new results would be precise and the simulation time reasonable.

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